Minor Project Synopsis Report

SMART DIGITAL WARDROBE

Project Category: University Based Project

Projexa Team Id-26E20741

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

in

CSE (Section -A)

to

K.R Mangalam University

By

***Lakshita Kalra** –2401010117*
***Vandana Sharma** – 2401010285*
***Sakshi Garg** – 2401010178*
***Saieena Negi** – 2401010081*
***Yuvraj Singh** – 2401010225*
***Anurag Upadhyay** – 2401010193*

Under the supervision of

Miss Mansi Kajal

Department of Computer Science and Engineering School of Engineering and Technology

K.R MANGALAM UNIVERSITY
Gurugram- 122001

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ABSTRACT

In the modern university environment, students manage a fast-paced lifestyle that demands efficiency, organization, and smart decision-making. One common yet often ignored challenge faced by students is effective wardrobe management. Students frequently experience difficulty in remembering what clothes they own, deciding appropriate outfits for different occasions, seasons, or academic events, and avoiding repetitive or unnecessary purchases. These issues lead to wasted time, increased expenses, and inefficient utilization of clothing resources.

The **Smart Digital Wardrobe** project presents a technology-driven solution to address this problem by providing a centralized digital platform where students can catalog, organize, and manage their wardrobe items virtually. The system allows users to store detailed information about each clothing item, such as category, color, type, season, and frequency of use. Through intelligent filtering and basic automation, the application assists users in selecting outfits efficiently and planning their wardrobe usage more effectively.

This project aligns with smart campus initiatives by integrating digital innovation into everyday student life. Additionally, it promotes sustainable fashion practices by encouraging reuse of existing clothes and minimizing unnecessary purchases. The Smart Digital Wardrobe thus addresses a real-world university-based problem by enhancing student convenience, improving personal organization, and supporting environmentally responsible behavior through technology.

KEYWORDS: Smart Wardrobe, Digital Inventory System, Student Lifestyle Management, Smart Campus Solutions, Sustainable Fashion

INTRODUCTION

University students today face multiple challenges related to academic responsibilities, extracurricular activities, social engagements, and personal management. Amid these responsibilities, personal organization—particularly wardrobe management—often becomes neglected. Students living in hostels or shared accommodations commonly struggle with limited space, poor organization, and lack of clarity about their available clothing options.

Choosing suitable outfits daily can lead to decision fatigue, wasted time, and stress, especially during early morning classes, presentations, interviews, or university events. Additionally, students may unintentionally purchase similar clothing items due to lack of awareness of what they already own, leading to financial inefficiency and wardrobe clutter.

With the rapid growth of digital technologies and smart campus concepts, there is a significant opportunity to apply technology to solve everyday lifestyle problems faced by students. The **Smart Digital Wardrobe** serves as a virtual wardrobe management system that digitizes clothing inventories and provides structured organization. By offering easy access to wardrobe data anytime and anywhere, the system enhances daily decision-making and improves students' overall lifestyle efficiency.

This project is designed as a university-based problem-solving initiative that focuses on student convenience, digital innovation, and sustainable practices within the campus ecosystem.

MOTIVATION

The motivation for developing the Smart Digital Wardrobe stems from common, real-life challenges experienced by students in their daily routines. Despite owning multiple clothing items, students often feel they have "nothing to wear" due to poor organization and lack of visibility into their wardrobe contents. This leads to frustration, poor time management, and unnecessary shopping.

Another major motivation is the growing concern around sustainability and responsible consumption. Fast fashion trends encourage frequent buying and disposal of clothes, which contributes to environmental degradation. Students, as responsible members of society, need tools that encourage mindful consumption and efficient utilization of resources.

Furthermore, the increasing adoption of digital platforms for task management, learning, and lifestyle organization highlights the need for similar solutions in personal management areas such as wardrobe planning. By creating a simple, student-friendly digital wardrobe system, this project aims to reduce daily stress, save time, promote sustainability, and enhance the overall quality of student life through smart technology.

LITERATURE REVIEW

The concept of digital wardrobe management has gained attention in recent years with the advancement of smart lifestyle applications and personalization technologies. Various studies and applications have explored the use of technology to assist users in managing clothing, fashion choices, and personal styling. However, most existing solutions focus on commercial fashion recommendations or e-commerce-driven personalization rather than addressing everyday wardrobe management problems faced by students.

Several mobile and web-based applications allow users to upload images of their clothing items and create a digital closet. While such applications offer convenience, they often lack structured categorization, usage tracking, and student-centric design. Moreover, many of these platforms require paid subscriptions or internet-intensive features, making them less accessible to university students.

Research in smart lifestyle management systems highlights the importance of digital inventories and personal data organization. Studies emphasize that visualizing owned resources helps users make informed decisions, reduces redundancy, and improves resource utilization. In the context of clothing management, digital inventories can significantly reduce impulsive buying behavior and promote sustainable fashion practices.

Recent advancements in artificial intelligence and machine learning have further enabled personalized recommendation systems. Some research proposes AI-based outfit suggestion engines that consider weather, occasion, and user preferences. However, these systems are often complex and
data-intensive, making them unsuitable for introductory-level university projects or basic student needs.

The Smart Digital Wardrobe project builds upon these research insights while maintaining simplicity and usability. It focuses on core wardrobe organization features, laying a strong foundation for future AI-based enhancements while remaining practical, accessible, and relevant to student life within the university ecosystem.

GAP ANALYSIS

A study of existing digital wardrobe and fashion management applications reveals several limitations. Many available solutions focus primarily on fashion recommendations or commercial e-commerce integration rather than personal wardrobe organization. These platforms are often complex, subscription- based, or designed for professional fashion use rather than everyday student needs.

Additionally, most existing systems do not consider the specific challenges faced by university students, such as limited storage space, tight budgets, and the need for quick outfit planning. There is also a lack of emphasis on sustainability and reuse of existing clothing within these applications.

The Smart Digital Wardrobe bridges these gaps by offering a simple, cost-effective, and student-centric solution. It focuses on essential wardrobe management features, ease of use, and campus relevance, making it suitable for daily student life. The system also lays the foundation for future enhancements such as AI-based outfit recommendations, further strengthening its practical value.

PROBLEM STATEMENT

Students often face difficulties in organizing and managing their personal wardrobes due to lack of structured information, limited time, and inefficient planning. This results in confusion during outfit selection, repetitive clothing purchases, financial inefficiency, and underutilization of existing wardrobe items.

There is a need for a digital system that enables students to catalog their clothing items, visualize their wardrobe contents, and plan outfits efficiently. Such a system should be easy to use, accessible, and aligned with student lifestyle requirements within the university ecosystem.

OBJECTIVES

The primary objectives of the Smart Digital Wardrobe project are as follows:

- To design and develop a digital platform for effective personal wardrobe management
- To help students organize clothing items based on category, color, season, and usage
- To assist students in daily outfit selection and planning.
- To reduce clothing wastage by encouraging reuse and mindful consumption
- To enhance time management and reduce decision fatigue among students
- To provide a user-friendly and scalable solution suitable for future enhancements

Tools/Technologies Used

The Smart Digital Wardrobe project uses modern web development tools to build a simple, efficient, and user-friendly system suitable for student use.

Frontend:
HTML is used to structure the web pages, CSS is used for styling and layout, and JavaScript is used to add interactivity such as form handling and filtering wardrobe items.

Backend:
Node.js with Express.js is used (if applicable) to handle server-side logic and manage communication between the frontend and database.

Database:
MongoDB is used to store wardrobe details such as clothing category, color, and season. For basic implementation, browser Local Storage can also be used.

Development Tools:
Visual Studio Code is used for writing and managing code, while GitHub is used for version control and team collaboration.

METHODOLOGY

The development of the Smart Digital Wardrobe follows a structured and systematic methodology to ensure efficiency, usability, and scalability of the system.

Requirement Analysis

This phase involves identifying user requirements through observation and discussion. The primary focus is to understand students' wardrobe management challenges, dressing preferences, and expectations from a digital wardrobe system.

Data Collection

In this stage, users upload clothing details such as category, color, and type into the system. Relevant external data, such as weather information, is also collected to support better outfit planning and future recommendation features.

System Design

The system architecture is designed by defining the frontend and backend structure. This includes designing the user interface, database schema, and data flow required to store and manage wardrobe items efficiently.

Implementation

Core functionalities are developed in this phase, including wardrobe item addition, categorization, filtering, and basic outfit planning. The wardrobe module and the foundation for a recommendation engine are implemented.

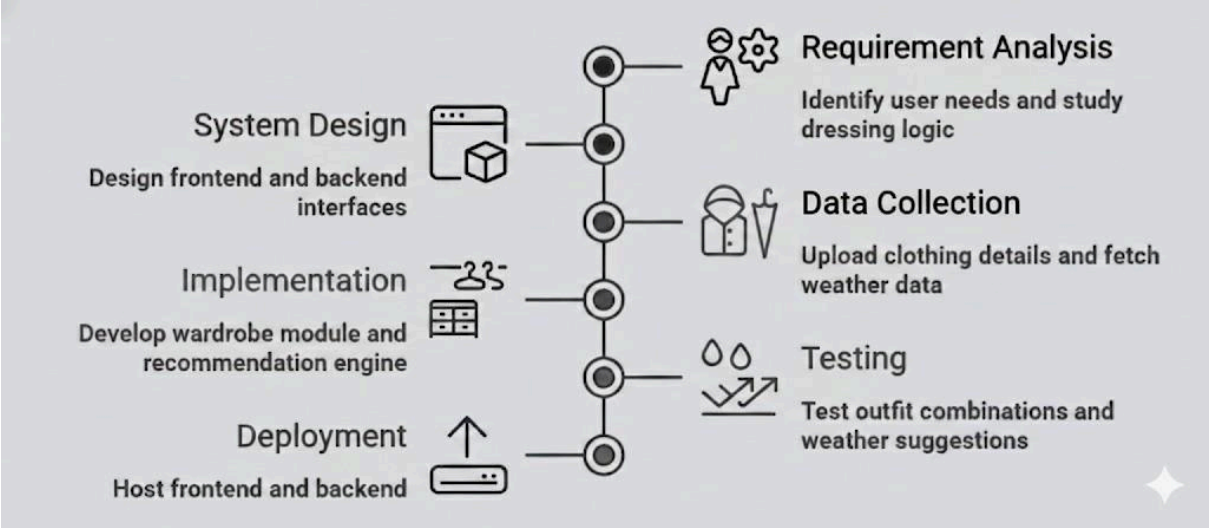
Testing

The system is tested using sample wardrobe data to verify functionality, accuracy, usability, and performance. Outfit combinations and weather-based suggestions are checked to ensure correct system behavior.

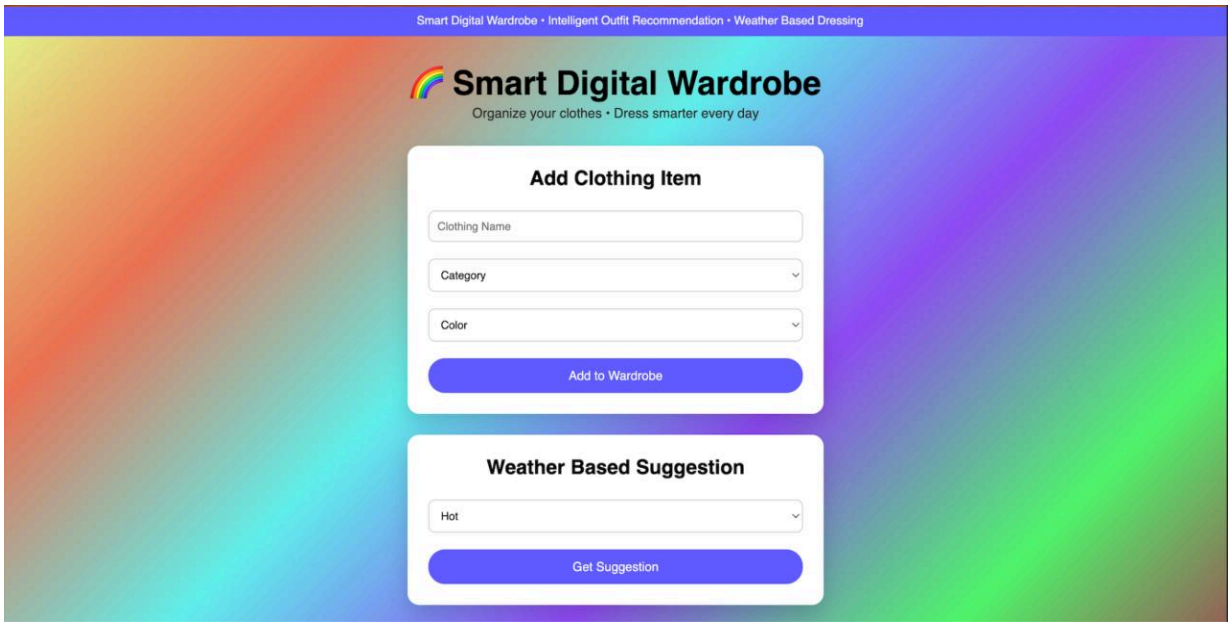
Deployment

The final system is deployed by hosting the frontend and backend components, making the Smart Digital Wardrobe accessible to users.

This methodology ensures a modular and scalable system design, allowing future integration of advanced features such as artificial intelligence–based outfit recommendations and usage analytics.



PROJECT SNAPSHOT



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